

RESEARCH

& INNOVATION UPDATES

Advanced Lightweight Prosthetic Hand

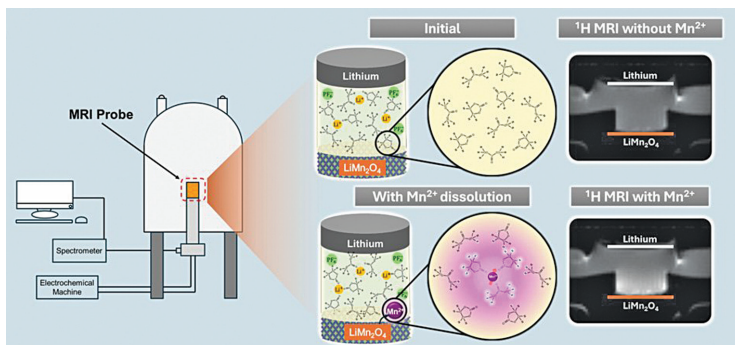


A team from the University of Science and Technology of China (USTC) has developed a lightweight prosthetic hand with 19 degrees of freedom (DOF), closely mimicking the human hand function. Using shape-memory alloys (SMA) as artificial muscles, it achieves high power-to-weight efficiency with a tendon-driven transmission system. The 0.37kg hand includes 23 sensors for precise control, a cooling module with 38 SMA actuators, and integrated voice recognition supporting 60 languages. It performs 33 traditional and six advanced grasping motions, enabling tasks like writing, playing chess, and using smartphones, revolutionising prosthetic technology.

Demonstration of the prosthetic hand's human-like gestures, gripping, operational functions, and patient experiments (Image by USTC; Credit: Hao Yang et al)

MRI Unlocks Lithium-Ion Battery Performance

Lithium-ion batteries power modern devices, but they degrade over time due to metal ion dissolution from the cathode into the electrolyte. Researchers at Tohoku University used MRI to detect this process with high sensitivity. Their study focused on Mn^{2+} dissolution from a $LiMn_2O_4$ cathode in a $LiPF_6$ EC:DMC electrolyte, confirming real-time metal ion movement. Testing an alternative $LiTFSI$ MCP electrolyte, they found it suppressed manganese dissolution. This breakthrough enables better battery material research, improving durability and performance for future energy storage solutions.



Schematic representation of the present work showing the increase in the MRI intensity confirming manganese dissolution from $LiMn_2O_4$ cathode (Credit: Hellar et al)